



assignment 6 , matlab

Linear Control Systems

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0.1 Question 4: Riderless Bicycle Stability Analysis

A riderless bicycle is stabilized by feedback from lean angle (output) to steering (input). The plant model is

$$G_p(s) = G_1(s)G_2(s) = \left[\frac{3.373}{s+8} \cdot \frac{1}{s^2-18.9} \right] \left[e^{-0.015s} \cdot \frac{(33.9)^2}{s^2+40.68s+(33.9)^2} \right], \quad (1)$$

and the PID controller is

$$G_c(s) = 2.5177 \left(1 + \frac{1.5431}{s} + 0.075s \right), \quad (2)$$

with **unity negative feedback**. The open-loop transfer function is

$$L(s) = G_c(s)G_p(s). \quad (3)$$

(a) Nyquist diagram of $L(s)$ and closed-loop stability using Nyquist criterion

Using MATLAB, plot the Nyquist diagram of the open-loop transfer function

$$L(s) = G_c(s)G_p(s)$$

(Hint: use pade to approximate the time delay or handle it directly with nyquist). Verify the closed-loop stability using the Nyquist stability criterion. Determine:

- the number of encirclements of the point -1 ,
- the number of open-loop unstable poles.

What the MATLAB code does:

1. **Define the plant factors** $G_1(s)$ and $G_2(s)$:

$$G_1(s) = \frac{3.373}{s+8} \cdot \frac{1}{s^2-18.9}, \quad G_{2,nodelay}(s) = \frac{\omega_n^2}{s^2+40.68s+\omega_n^2}, \quad \omega_n = 33.9.$$

2. **Approximate the time delay** $e^{-0.015s}$ by a Padé model of order $N_{pade} = 4$:

$$e^{-T_d s} \approx \frac{num_D(s)}{den_D(s)}, \quad T_d = 0.015 \text{ s},$$

implemented by `[numD,denD]=pade(Tdelay,N_pade)` and `DelayApprox=tf(numD,denD)`. Then the delayed block is built as

$$G_2(s) = \left(\frac{num_D(s)}{den_D(s)} \right) G_{2,nodelay}(s).$$

3. **Define the PID controller:**

$$G_c(s) = 2.5177 \left(1 + \frac{1.5431}{s} + 0.075s \right),$$

where the $1s$ term introduces an integrator (pole at the origin).

4. **Form the open-loop transfer:**

$$L(s) = G_c(s)G_p(s) = G_c(s)G_1(s)G_2(s).$$

5. **Plot the Nyquist diagram** using `nyquist(L)` and mark the critical point -1 .

6. **Count open-loop unstable poles** by computing `pL=pole(L)` and then counting how many poles satisfy $\Re\{p\} > 0$.

7. **Estimate the encirclement count** around -1 from sampled frequency response data:

- build $L(j\omega)$ over a dense log-spaced frequency grid $\omega \in [10^{-4}, 10^4]$,
- mirror it to emulate the negative-frequency Nyquist branch,
- shift by $+1$ so encirclements of -1 become encirclements of 0:

$$z(\omega) = L(j\omega) + 1,$$

- compute the net winding number using the unwrapped angle change:

$$N \approx \frac{\Delta \angle z}{2\pi}.$$

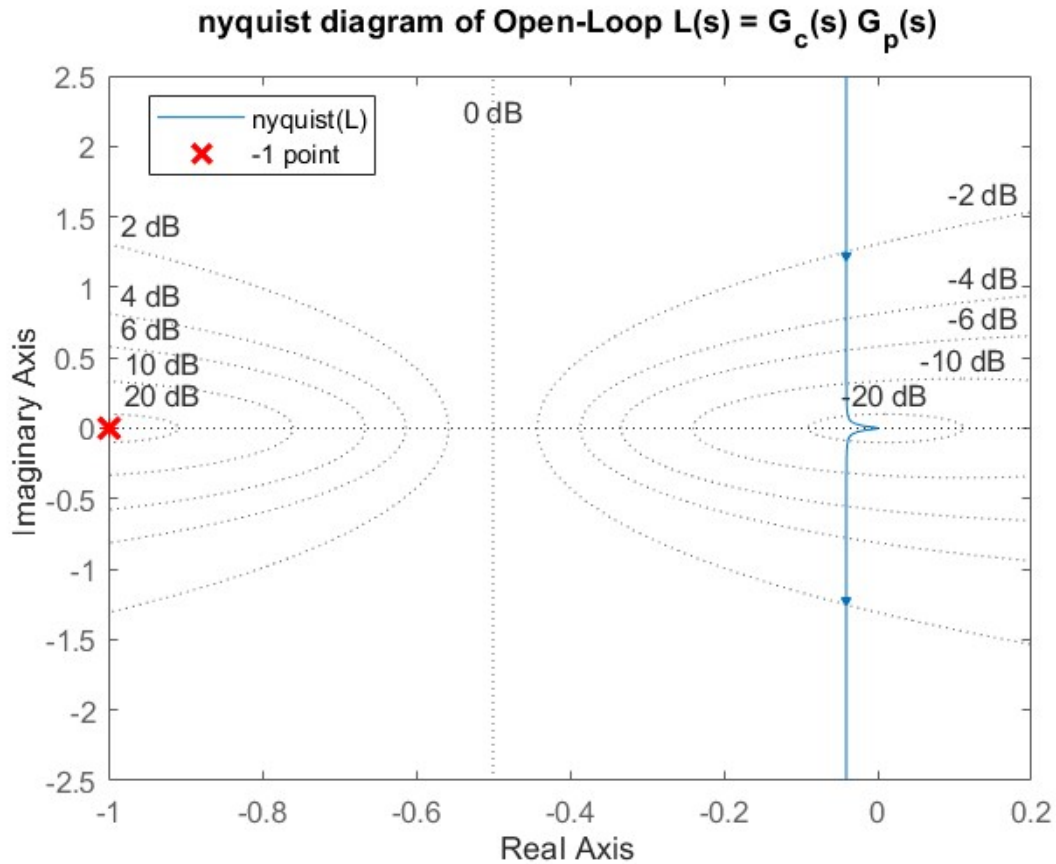


Figure 1: Nyquist plot of $L(s) = G_c(s)G_p(s)$ (delay approximated by Padé order 4). The critical point -1 is marked on the real axis.

Command Window results (open-loop poles and encirclement count). The script prints the open-loop poles and then reports:

- **Number of open-loop unstable poles:** Count: RHP poles = 1. (So $L(s)$ has exactly one pole in the right-half plane.)
- **Pole at the origin:** Count: poles at origin = 1. (This comes from the integrator term $1/s$ in the PID.)
- **Nyquist encirclements of -1 :** Estimated Nyquist encirclements of -1 (N) = 0.

Nyquist stability test (using only $N, N_{\text{cup}}, N_{\text{oup}}$). For unity negative feedback, the closed-loop characteristic equation is

$$1 + L(s) = 0.$$

The Nyquist criterion states (in the course notation) that:

The net encirclement count of the critical point -1 by the Nyquist plot of $L(j\omega)$, denoted by

$$N = N_{\text{cup}} = N_{\text{oup}},$$

must satisfy the required encirclement condition imposed by the number of open-loop unstable poles.

Here, the open-loop has **one** unstable pole (RHP pole count = 1), but the computed encirclement count is

$$N = N_{\text{cup}} = N_{\text{oup}} = 0.$$

Therefore the Nyquist requirement for closed-loop stability is **not satisfied**, and the closed-loop is **unstable**. This matches MATLAB's output warning: *"The closed-loop system is unstable."*

Important remark about the pole at $s = 0$. Because $L(s)$ contains an integrator, the Nyquist contour passes through a singularity at $\omega = 0$. In rigorous Nyquist analysis one uses a small indentation around the origin. Numerically, the script avoids evaluating exactly at $\omega = 0$ by using a frequency grid starting at $\omega = 10^{-4}$, which is consistent with the practical Nyquist plotting approach in MATLAB. “

(b) Bode magnitude of loop gain and gain crossover frequency ω_c

Plot the magnitude $|L(j\omega)|$ versus frequency and determine the gain crossover frequency ω_c such that $|L(j\omega_c)| = 1$.

The MATLAB section labeled % Partb of Q4:

- Plots `bodemag(L)`.
- Uses `margin(L)` to compute the gain crossover frequency `Wcp`, which MATLAB defines as the frequency where $|L(j\omega)| = 1$.
- Verifies by directly evaluating $|L(jWcp)|$ via `freqresp`.

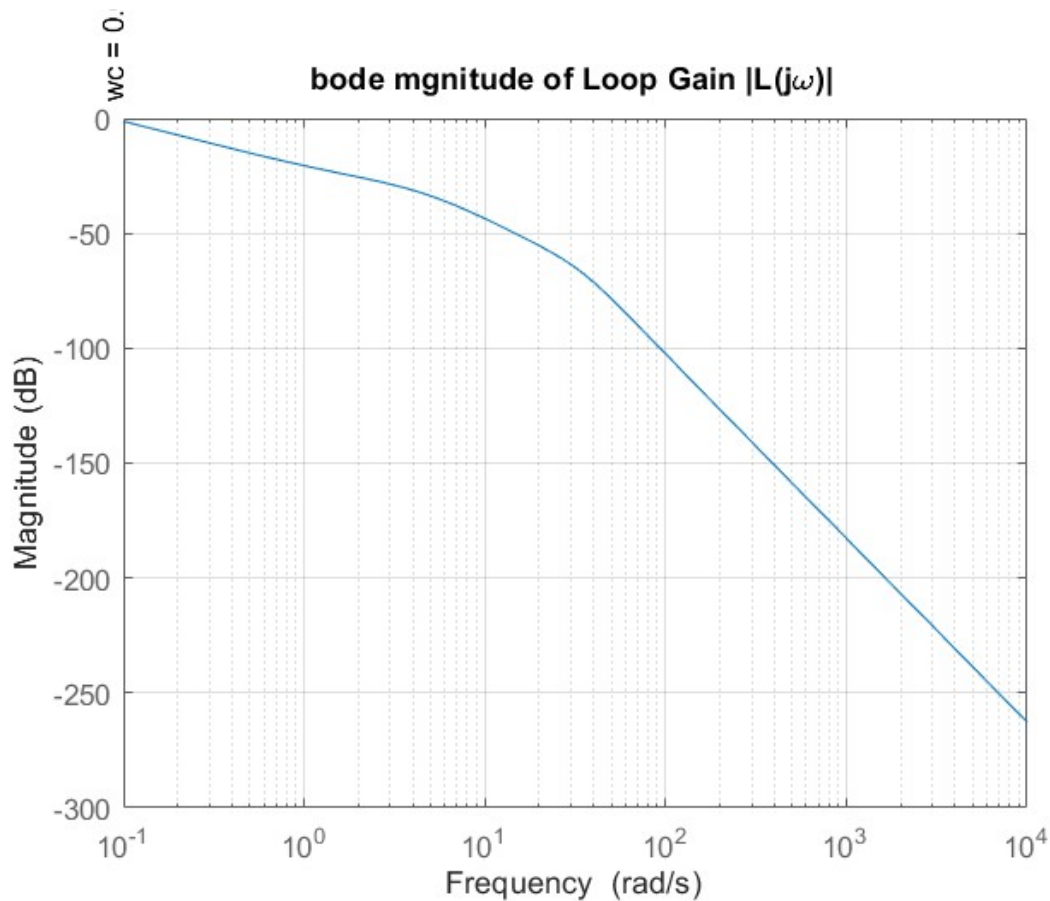


Figure 2: Bode magnitude plot of the loop gain $|L(j\omega)|$ and indication of the gain crossover frequency ω_c .

Command Window results (gain crossover). The output reports:

$$\omega_c = \text{Wcp} = 0.0867345 \text{ rad/s},$$

and the code verifies:

$$|L(j\omega_c)| \approx 1.$$

The margin function also reports a **negative phase margin** ($\text{PM} = -87.65^\circ$), which is consistent with the unstable closed-loop conclusion from part (a).

(c) Low- and high-frequency magnitude bounds relative to ω_c

Task statement. From the magnitude plot, find:

- the minimum value of $|L(j\omega)|$ for $\omega \leq 0.1\omega_c$,
- the maximum value of $|L(j\omega)|$ for $\omega \geq 10\omega_c$.

Code location (what part answers this). The MATLAB section labeled % part c of Q4:

- Sets $\omega_c = \omega_{cp}$.
- Builds a log-spaced grid for the **low-frequency** range $[\omega_{\min}, 0.1\omega_c]$ and computes $|L(j\omega)|$ via `freqresp` to find the minimum magnitude.
- Builds a log-spaced grid for the **high-frequency** range $[10\omega_c, \omega_{\max}]$ (with a cap) and finds the maximum magnitude.

Command Window numerical answers. Using $\omega_c = 0.0867345$ rad/s:

- Low-frequency region $\omega \leq 0.1\omega_c$ (i.e., $\omega \leq 0.00867345$):

$$\min_{\omega \in [10^{-6}, 0.1\omega_c]} |L(j\omega)| = 9.9925 \quad \text{at } \omega = 0.00867345 \text{ rad/s.}$$

- High-frequency region $\omega \geq 10\omega_c$ (i.e., $\omega \geq 0.867345$):

$$\max_{\omega \in [10\omega_c, 867.345]} |L(j\omega)| = 0.106585 \quad \text{at } \omega = 0.867345 \text{ rad/s.}$$

Interpretation. These results indicate that:

- For very low frequencies (up to $0.1\omega_c$), the loop gain magnitude is *well above* unity (minimum ≈ 10), implying strong low-frequency loop amplification.
- For high frequencies (beyond $10\omega_c$), the loop gain is *well below* unity (maximum ≈ 0.107), implying the loop is effectively open at high frequency (good noise attenuation behavior).

Despite these favorable magnitude trends, the Nyquist result in part (a) shows that the closed-loop is unstable due to the combination of an open-loop RHP pole ($P = 1$) and the observed encirclement count $N = 0$, producing $Z = 1$ RHP closed-loop pole.

0.2 Question 5: Rational approximations of the time delay e^{-sT} (with $T = 1.5$)

Approximating a pure delay by a rational transfer function is useful because standard frequency-domain and time-domain analysis tools (Bode, Nyquist, root-locus, step simulation) require finite-dimensional LTI models. In this question two approaches are considered:

- **Split-Taylor (not Pade):**

$$e^{-sT} = \frac{e^{-sT/2}}{e^{+sT/2}} \quad (4)$$

and each exponential term is approximated by a truncated Taylor series.

- **Padé:**

$$e^{-sT} \approx R_{m,n}(Ts) = \frac{\sum_{i=0}^m p_i(Ts)^i}{\sum_{i=0}^n q_i(Ts)^i}, \quad p_0 = q_0 = 1. \quad (5)$$

(a) Bode diagrams of Split-Taylor approximations ($n = 1, 2, 3, 4, 5$)

Using MATLAB with $T = 1.5$, compare the Bode diagrams of Taylor-series-based rational approximations of orders $n = 1, \dots, 5$ on the same figure.

The provided MATLAB code under:

```
for idx = 1:numel(n_list) ... end and the subsequent bodeplot(...) loop
```

constructs the Split-Taylor rational model

$$\widehat{G}_n(s) = \frac{\sum_{k=0}^n \frac{(-T/2)^k}{k!} s^k}{\sum_{k=0}^n \frac{(+T/2)^k}{k!} s^k}, \quad (6)$$

implemented by building ascending coefficients for the truncated series in s and then flipping them to descending order for `tf(num,den)`.

Command Window output interpretation (models). The printed results match the expected structure:

$$n=1: \widehat{G}_1(s) = \frac{-0.75s+1}{0.75s+1},$$

$$n=2: \widehat{G}_2(s) = \frac{0.2812s^2-0.75s+1}{0.2812s^2+0.75s+1},$$

$$n=3: \widehat{G}_3(s) = \frac{-0.07031s^3+0.2812s^2-0.75s+1}{0.07031s^3+0.2812s^2+0.75s+1}, \text{ etc.}$$

The numerator coefficients alternate in sign, while the denominator coefficients are all positive, consistent with the split $\frac{e^{-sT/2}}{e^{+sT/2}}$.

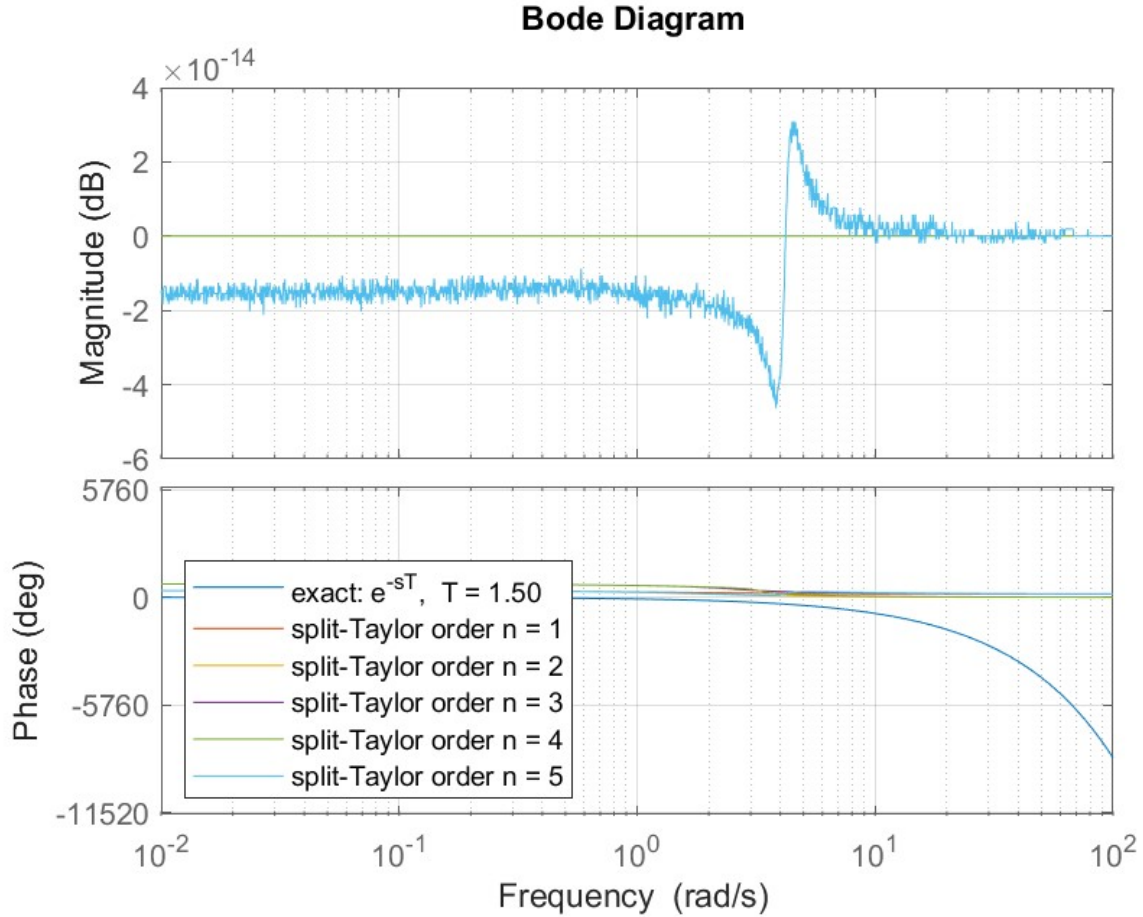


Figure 3: Bode magnitude/phase: exact delay e^{-sT} vs Split-Taylor approximations ($n = 1..5$, $T = 1.5$).

At low frequencies ($\omega T \ll 1$) all approximations closely match the exact delay because their series expansion matches the first terms of e^{-sT} . As frequency increases, both magnitude and phase deviate: the rational models cannot reproduce the exact constant-magnitude delay response $|e^{-j\omega T}| = 1$ for all ω , and phase errors grow with ω . Higher order generally improves the match over a wider frequency band, but can also introduce undesirable pole/zero dynamics.

(b) Step responses of Split-Taylor approximations; stability and initial behavior

Compare the step responses of the Taylor approximations and comment on stability and the initial behavior.

The added MATLAB section labeled:

```
Part (b) Step responses ... and the diagnostic loop printing isstable, poles, and
y(0+)
```

simulates step responses (same time grid for all models) and checks stability by evaluating pole locations.

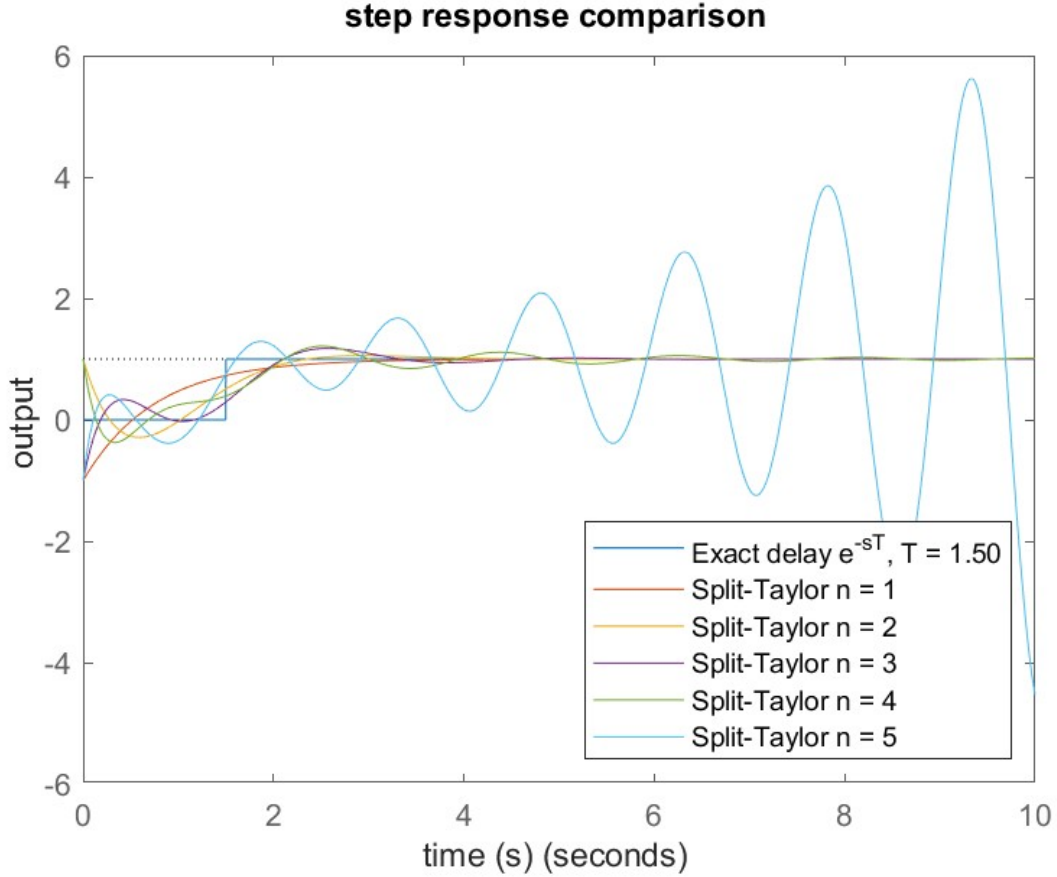


Figure 4: Step responses: exact delay e^{-sT} vs Split-Taylor approximations ($n = 1..5$, $T = 1.5$).

Interpretation of Command Window results (stability). From the printed pole sets:

- For $n = 1, 2, 3, 4$ all poles have negative real part, therefore `isstable=1` and the step responses remain bounded.
- For $n = 5$ the code reports `isstable=0` because a complex pole pair has positive real part

$$0.3197 \pm 4.1711j,$$

so the step response grows unbounded (unstable approximation).

Initial behavior and “pre-response” (non-causality). The true delay has $y(t) = 0$ for $t < T$ under a unit step input and then begins rising at $t = T = 1.5$ s. In contrast, rational delay approximations can exhibit *pre-response*: the output starts changing before $t = T$. This is a well-known artifact of approximating a non-rational, non-local operator (delay) by a finite-dimensional rational system.

Sign of $y(0^+)$ in the simulation output. The reported values $\hat{y}(0^+) \approx \pm 1$ alternating with n arise from the specific numerator/denominator structure of the Split-Taylor rational functions and the resulting high-frequency (feedthrough-like) behavior of those models. The key practical point is: *the initial value can be non-physical and even change sign depending on the approximation order*. For analysis/design, the important checks are pole stability and qualitative resemblance near $t \approx T$.

(c) Derive Pade coefficients for $R_{2,2}(Ts)$ and compare to Taylor up to order 4

Derive coefficients p_i, q_i for the Pade approximation $R_{2,2}(Ts)$ and determine whether they are identical to Taylor-series coefficients up to order 4.

The MATLAB block labeled:

Part (c) Derive the coefficients p_i, q_i for the Padé approximation $R_{2,2}(Ts)$ implements the given closed-form expressions:

$$p_i = (-1)^i \frac{(m+n-i)!m!}{(m+n)!i!(m-i)!}, \quad q_i = \frac{(m+n-i)!n!}{(m+n)!i!(n-i)!}, \quad (7)$$

with $m = n = 2$.

Derived coefficients (as printed). For $m = n = 2$:

$$p_0 = 1, \quad p_1 = -12, \quad p_2 = 112, \quad q_0 = 1, \quad q_1 = +12, \quad q_2 = 112.$$

With $Ts = Ts$ and $T = 1.5$ the resulting rational approximation becomes:

$$R_{2,2}(Ts) = \frac{0.1875s^2 - 0.75s + 1}{0.1875s^2 + 0.75s + 1},$$

which matches the MATLAB printout.

Are Padé coefficients identical to Taylor coefficients up to order 4? *No, the coefficient sets (p_i, q_i) are not the same thing as Taylor coefficients of e^{-x} .* What Padé guarantees is that the *Maclaurin series of the rational function* matches the Taylor series of the exponential up to order $(m+n)$:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots, \quad x = Ts.$$

The MATLAB output confirms that the series coefficients of $R_{2,2}$ expanded about $x = 0$ match e^{-x} up to x^4 exactly (flag =1 in the comparison). Therefore:

$$R_{2,2}(x) = e^{-x} + O(x^5), \quad (8)$$

which is the defining property of Padé with $(m, n) = (2, 2)$.

(d) Bode diagrams of balanced Padé approximations $R_{1,1}$ – $R_{4,4}$

Using MATLAB with $T = 1.5$, compare Bode diagrams of balanced Pade approximations $R_{1,1}, R_{2,2}, R_{3,3}, R_{4,4}$.

The MATLAB block labeled:

Part (d) Bode comparison: balanced Pade ...

uses Control System Toolbox `pade(T,N)` to generate $R_{N,N}$ and overlays all Bode plots together with the exact delay.

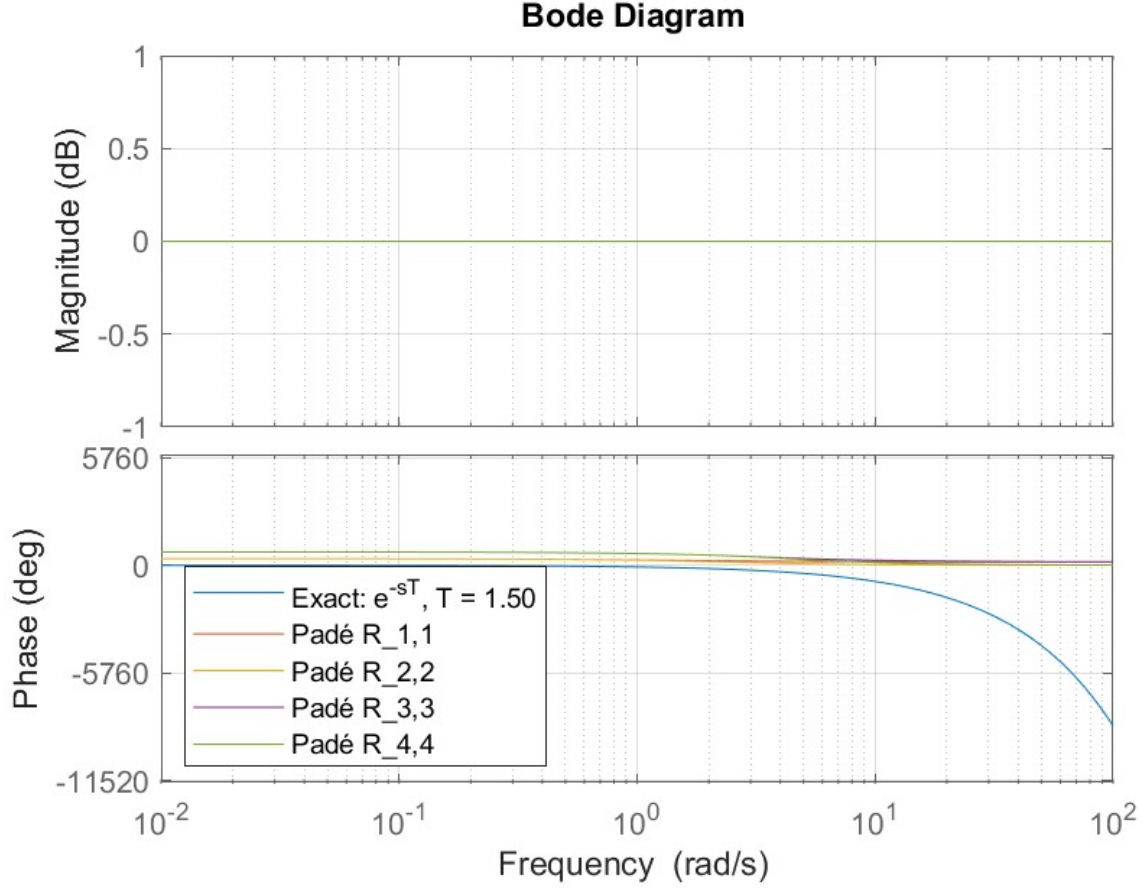


Figure 5: Bode comparison: exact delay e^{-sT} vs balanced Padé approximations $R_{1,1}$, $R_{2,2}$, $R_{3,3}$, $R_{4,4}$ for $T = 1.5$.

Stability result (from Command Window). The printed check shows:

$$R_{1,1}, R_{2,2}, R_{3,3}, R_{4,4} \text{ are all stable (isstable=1)}.$$

Hence their frequency responses are well-defined and do not exhibit divergent time-domain behavior due to unstable poles.

Bode-plot interpretation. Balanced Padé approximations are designed to match the delay response around low-to-moderate frequencies, producing improved phase/magnitude accuracy over a wider band as N increases. Unlike the exact delay (unit magnitude for all ω), rational Padé models deviate at higher frequencies, but higher order generally postpones that deviation. These approximations are therefore widely used to approximate phase lag in loop-shaping and margin computations.

(e) Balanced Padé step responses and discontinuity at $t = 0$

Key observation (from step-response behavior). Equal-order Padé $R_{n,n}$ typically shows an **instantaneous jump at $t = 0^+$** under a unit-step input, unlike the true delay which remains zero until $t = T$.

Reason (direct feedthrough / non-zero high-frequency gain). For $m = n$, the rational function is *proper*

but not strictly proper, so:

$$\lim_{s \rightarrow \infty} R_{n,n}(s) = \frac{\text{leading numerator coefficient}}{\text{leading denominator coefficient}} \neq 0. \quad (9)$$

A non-zero limit implies a direct feedthrough term in time domain, producing a discontinuity at $t = 0^+$ for step inputs. This is a known artifact of equal-order Padé approximations and explains the observed jump.

(f) Unbalanced Padé: compare $R_{1,2}$, $R_{2,3}$, $R_{3,4}$ (Bode and step)

How to compare using existing results. This comparison is conceptual and relies on the same principles visible in the earlier plots: frequency-domain fit vs time-domain realism.

Main differences. For $m < n$ the approximation is **strictly proper**, hence:

$$\lim_{s \rightarrow \infty} R_{m,n}(s) = 0. \quad (10)$$

Therefore unbalanced Padé does **not** have direct feedthrough and the step response does **not** jump at $t = 0^+$; it starts smoothly from zero, which is more physically plausible for a delay approximation. However, because the numerator order is lower, high-frequency phase/magnitude matching is typically worse than balanced Padé, so the Bode match is usually less accurate near the upper end of the plotted band.

(g) Why balanced Padé is preferred in frequency domain, while $m < n$ is preferred in time domain

Frequency-domain analysis (loop-shaping, margins). Balanced Padé ($m = n$) provides the **best phase and magnitude match** around low-to-mid frequencies for a given approximation order, making it preferred for Bode/Nyquist-based stability-margin calculations. The non-physical step discontinuity is irrelevant in pure frequency-domain design tasks.

Time-domain simulation (transient response). In time-domain simulations, a direct feedthrough (present in $m = n$) creates an artificial instantaneous jump that is not representative of a real delay element. Using $m < n$ makes the model strictly proper, avoids the $t = 0$ discontinuity, and yields smoother, more realistic transient behavior. Hence **unbalanced Padé is often chosen for simulations**, while **balanced Padé is preferred for frequency-domain analysis**.

Question 6 :

Plant and controller:

$$G(s) = \frac{10}{(s+1)(s^2+2s+4)}, \quad G_c(s) = K \frac{s+2}{s+1},$$

unity negative feedback. Define the loop gain

$$L(s) = G_c(s)G(s) = \frac{10K(s+2)}{(s+1)^2(s^2+2s+4)}.$$

Closed-loop transfer from reference to output:

$$T(s) = \frac{L(s)}{1 + L(s)}.$$

a)

For unity feedback and a unit step input $r(t) = 1(t)$, the error transfer is

$$\frac{E(s)}{R(s)} = \frac{1}{1 + L(s)}.$$

Using the Final Value Theorem (closed-loop stable):

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{1}{1 + L(s)} = \frac{1}{1 + L(0)}.$$

Compute the DC loop gain:

$$L(0) = \frac{10K(0+2)}{(0+1)^2(0^2+2 \cdot 0+4)} = \frac{20K}{4} = 5K.$$

Hence,

$$e_{ss}(K) = \frac{1}{1 + 5K}.$$

(Interpretation: the system is Type-0 (no integrator), so the step error is finite and controlled by the position error constant $K_p = L(0) = 5K$.)

b)

$$e_{ss}(20) = \frac{1}{1 + 5(20)} = \frac{1}{101} \approx 0.00990.$$

As a percent of the unit step amplitude:

$$100e_{ss} \approx 0.99\% < 9\%.$$

$$e_{ss}(20) \approx 0.00990 (< 0.09).$$

c)

With $K = 20$:

$$L(s) = \frac{200(s+2)}{(s+1)^2(s^2+2s+4)}.$$

On the Nichols chart, you read:

- Peak closed-loop magnitude (resonant peak) M_p in dB (the maximum of $|T(j\omega)|$ mapped via Nichols M-contours),
- Phase margin P.M. from the loop curve relative to the -180° crossing at 0 dB.

Practically, we compute the same quantities via frequency response:

$$M_p \equiv \max_{\omega} |T(j\omega)|, \quad M_{p,\text{dB}} = 20 \log_{10}(M_p).$$

Estimating ζ from M_p . If the dominant dynamics are well-approximated by a 2nd-order underdamped mode, then the resonant peak satisfies (for $\zeta < 1/\sqrt{2}$):

$$M_p \approx \frac{1}{2\zeta \sqrt{1 - \zeta^2}}.$$

Given M_p (linear, not dB), solve this equation for ζ (numerically is easiest).

Percent overshoot from ζ . For a standard 2nd-order underdamped step response,

$$\%OS \approx 100 \exp\left(-\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}\right).$$

Note: The MATLAB code provided in zip file that i uploaded computes $M_{p,\text{dB}}$, P.M., then solves for ζ and estimates %OS. (becuase this part included l..matlab at some parts i decided to all of it in latex , anwering one part of the same question in papaer in another pdf and next part that incldued matlab here in this pdf would make sense) command window result : Part (c): Nichols-based quantities $M_p = 1.228$ dB Phase Margin = -68.96 deg Estimated zeta = 0.502

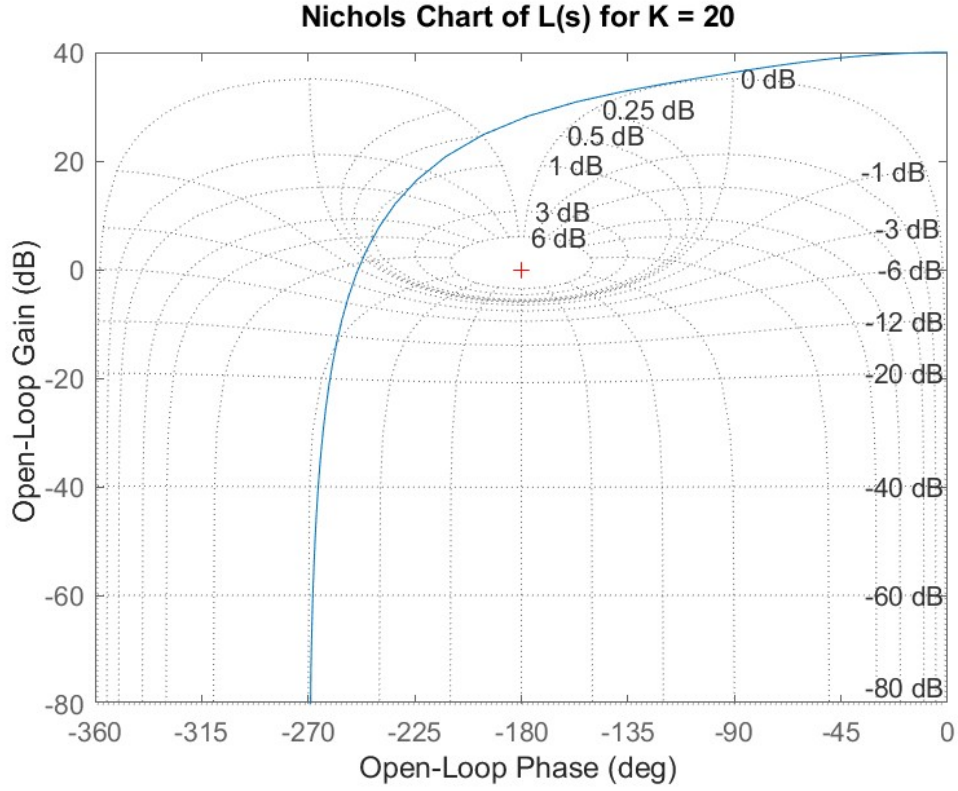


Figure 6: Nichols chart of $L(s)$ for $K = 20$.

d)

Given the design target

$$M_{p,\text{dB}} \approx 2.67 \text{ dB} \implies M_p \approx 10^{(2.67/20)} \approx 1.36.$$

On a Nichols chart, changing the scalar gain K shifts the loop curve *vertically* by

$$\Delta(\text{dB}) = 20 \log_{10} \left(\frac{K_{\text{new}}}{K_{\text{old}}} \right),$$

because magnitude changes while phase stays the same. So “shifting downward” means reducing K .

Using the MATLAB search in the code (matching the target $M_{p,\text{dB}}$), we obtain approximately

$$K_{\text{new}} \approx 6.23 \quad (\text{gives } M_{p,\text{dB}} \approx 2.67 \text{ dB}).$$

Then the corresponding steady-state error is

$$e_{ss}(K_{\text{new}}) = \frac{1}{1 + 5K_{\text{new}}} \approx \frac{1}{1 + 5(6.23)} \approx 0.0311.$$

$$e_{ss}(6.23) \approx 3.11\%.$$

commandd window result : Part (d): Gain retuning New K 6.2308 ess 0.0311 (3.11

e)

We generate $G(s)$, $G_c(s)$, form $T(s) = \frac{L}{1+L}$, plot the step response, then use `stepinfo` for overshoot, settling time, and peak time. The steady-state error is computed from the DC gain:

$$y(\infty) = \lim_{s \rightarrow 0} T(s) = T(0) = \text{dcgain}(T), \quad e_{ss} = 1 - y(\infty).$$

(f) For-loop table for $K = [4.44, 10, 20]$ and comparison with Nichols expectations

We run a loop over K values, store:

$$\%OS, \quad T_s, \quad T_p, \quad e_{ss},$$

and optionally add $M_{p,\text{dB}}$ and P.M. for comparison.

g)

Phase margin is extracted from `margin(L)` (loop gain). Bandwidth ω_B is the closed-loop -3 dB bandwidth of $T(s)$ (via `bandwidth(T)`).

To validate settling time:

$$T_s \approx \frac{4}{\zeta \omega_n}, \quad \omega_B \approx (-1.19\zeta + 1.85) \omega_n.$$

So,

$$\omega_n \approx \frac{\omega_B}{-1.19\zeta + 1.85}, \quad T_s^{(approx)} \approx \frac{4}{\zeta \omega_n}.$$

We estimate ζ from M_p (as in part (c)), then compute ω_n from ω_B , and finally compare $T_s^{(approx)}$ to MATLAB's `stepinfo` settling time.

g)

Overlay the three step responses on one figure and observe:

- Increasing K decreases $e_{ss} = \frac{1}{1+5K}$ (better tracking).
- Transient behavior (overshoot/peaking, speed) changes; Nichols/loop margins explain the trade-off.

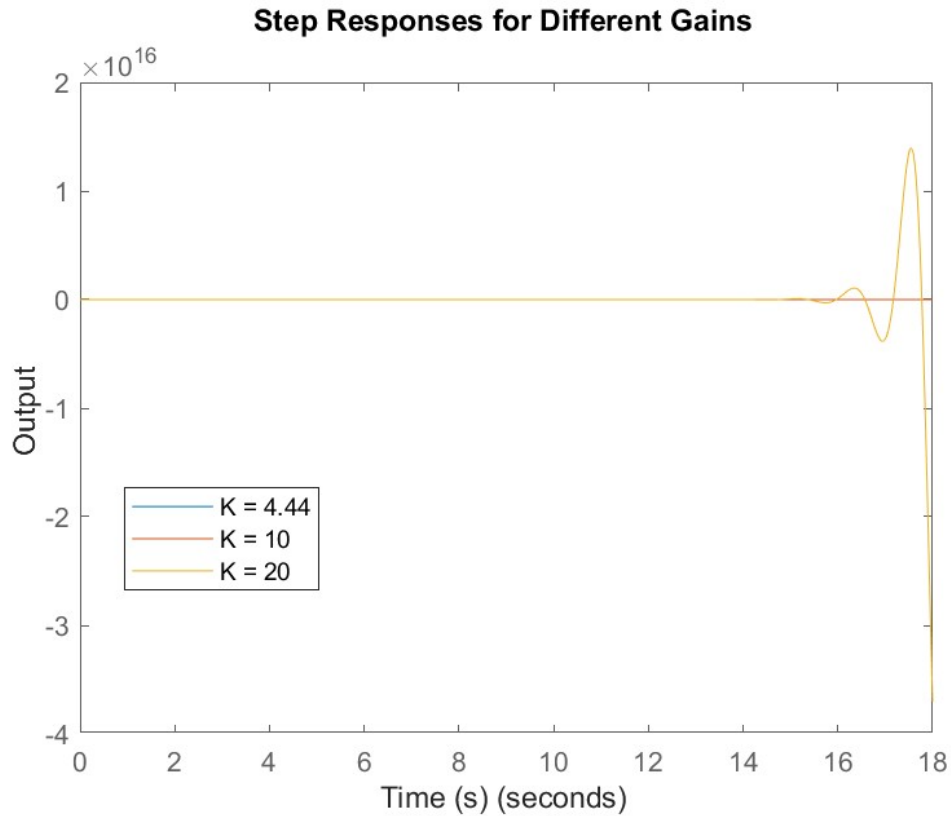


Figure 7: Overlaid step responses for $K = 4.44, 10, 20$

QUESTION 7

$$L(s) = K \frac{e^{-Ts}}{s+P}, \quad \text{unity negative feedback.}$$

Time delay makes the system infinite-dimensional, but the Nyquist criterion still applies to the loop frequency response $L(j\omega)$.

(a) Conditions for the Nyquist plot to pass through the critical point -1

Marginal stability (boundary of stability) occurs when the Nyquist curve passes exactly through the critical point $-1 + j0$. That means:

$$L(j\omega^*) = -1.$$

Write this as magnitude and phase conditions:

$$|L(j\omega^*)| = 1, \quad \angle L(j\omega^*) = (2\ell + 1)\pi, \ell \in \mathbb{Z}.$$

Now compute $L(j\omega)$:

$$L(j\omega) = K \frac{e^{-j\omega T}}{P + j\omega}.$$

Magnitude condition.

$$|L(j\omega)| = K \frac{|e^{-j\omega T}|}{|P + j\omega|} = K \frac{1}{\sqrt{P^2 + \omega^2}}.$$

So $|L(j\omega^*)| = 1$ implies

$$K = \sqrt{P^2 + (\omega^*)^2}.$$

Phase condition.

$$\angle L(j\omega) = \angle K + \angle e^{-j\omega T} - \angle(P + j\omega).$$

For $K > 0$, $\angle K = 0$. Also $\angle e^{-j\omega T} = -\omega T$, and

$$\angle(P + j\omega) = \tan^{-1}\left(\frac{\omega}{P}\right).$$

Hence

$$\angle L(j\omega) = -\omega T - \tan^{-1}\left(\frac{\omega}{P}\right).$$

The crossing condition $\angle L(j\omega^*) = (2\ell + 1)\pi$ is therefore

$$-\omega^* T - \tan^{-1}\left(\frac{\omega^*}{P}\right) = (2\ell + 1)\pi.$$

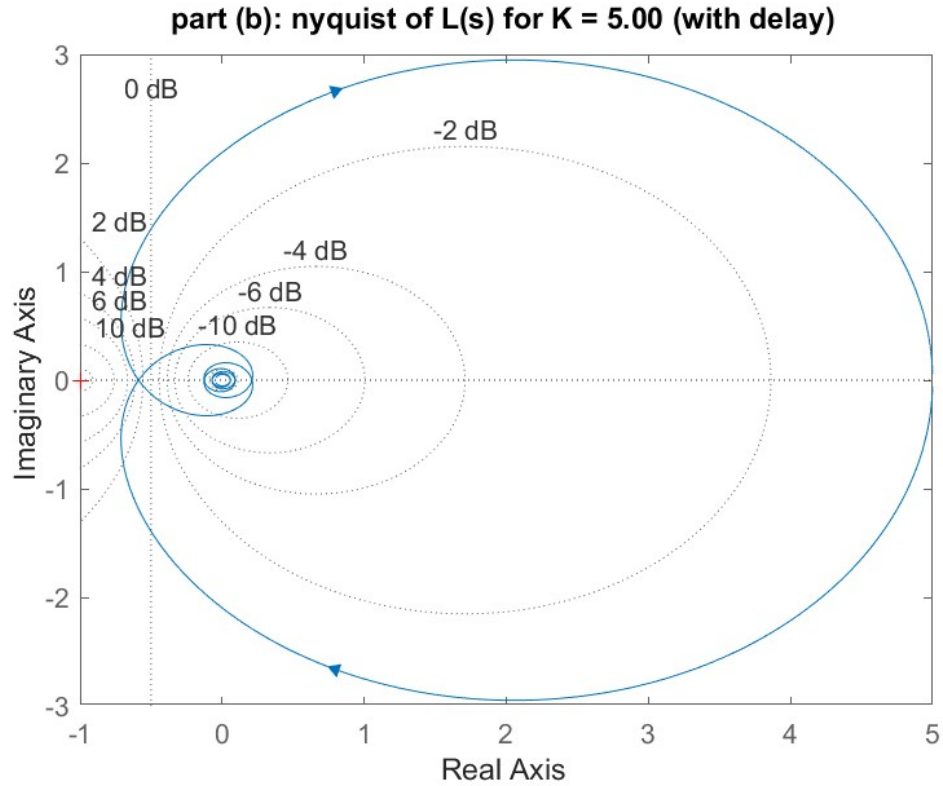
Equivalently (using the principal $-\pi$ crossing for the *first* onset of instability),

$$\omega^* T + \tan^{-1}\left(\frac{\omega^*}{P}\right) = \pi.$$

This pair (phase + magnitude) gives the marginal stability boundary (ω^*, K) .

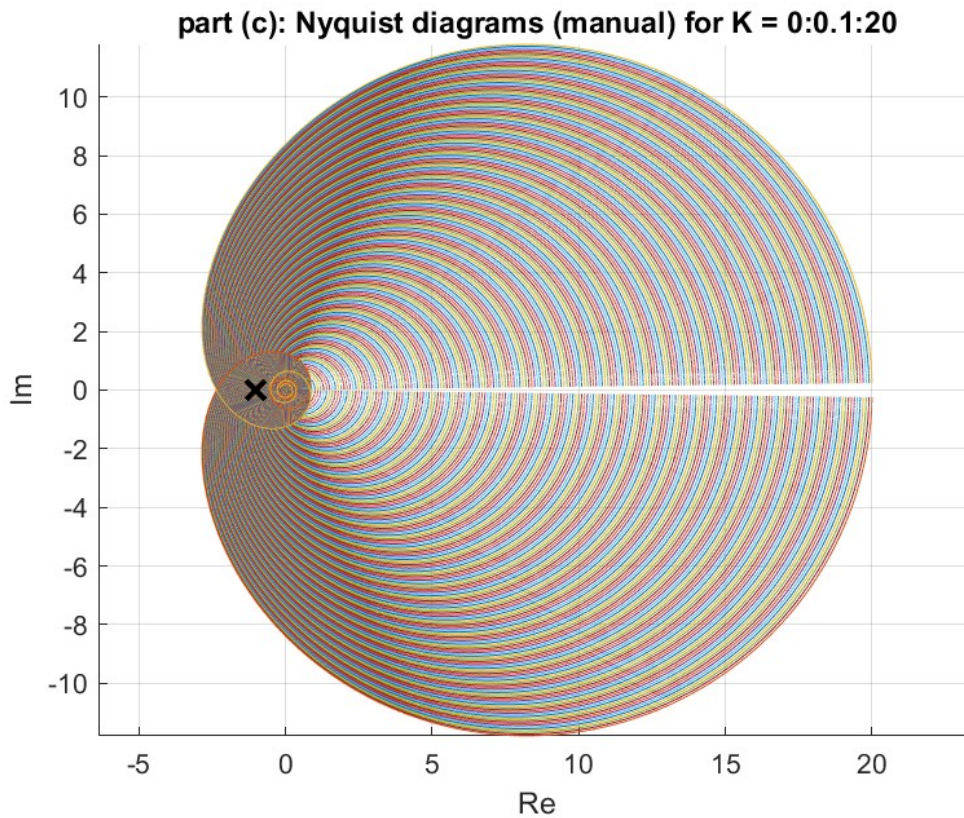
(b) MATLAB Nyquist plot for a given K (with $T = 0.2, P = 1$)

Use `tf` with 'InputDelay', T . The Nyquist command supports delays directly; alternatively, Padé can be used, but direct delay is preferred here.



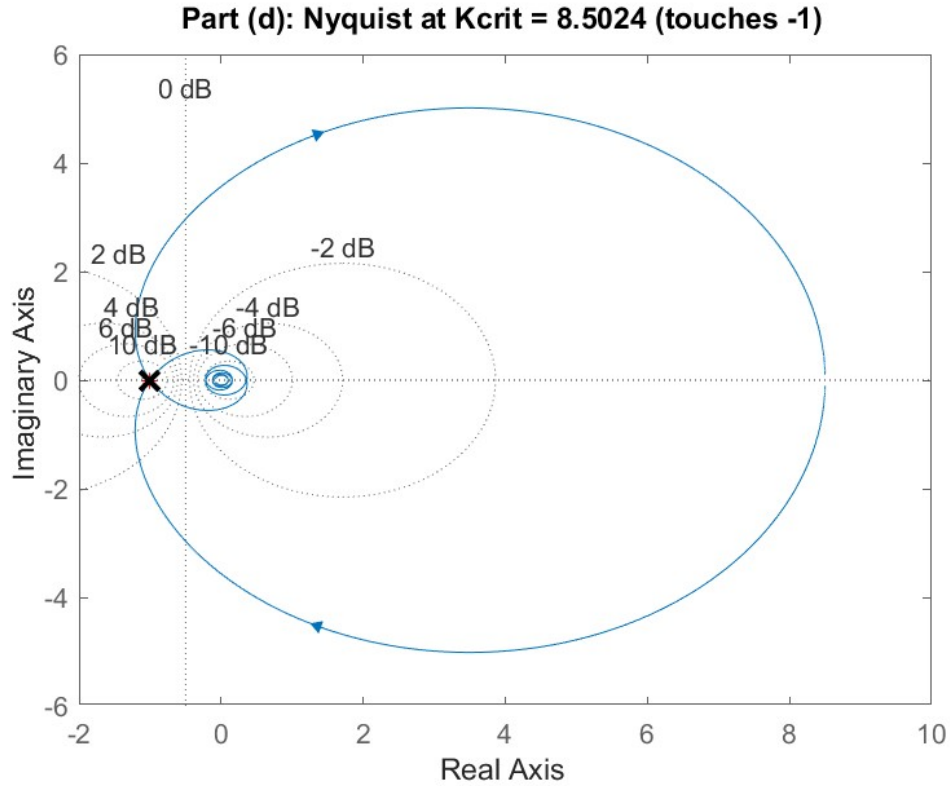
(c) Sweep $K \in [0, 20]$ and observe scaling

For fixed T, P , multiplying K scales the Nyquist curve radially (magnitudes scale by K); the delay contributes frequency-dependent phase lag, producing an inward spiral-like behavior as ω increases.



(d) Sketch and interpretation at the touching case

At the critical K , the Nyquist curve touches -1 at ω^* . Because the phase keeps decreasing with ω (due to $-\omega T$), the curve can spiral and may cross the negative real axis multiple times for larger ω , potentially enabling multiple near-crossings.



(e) Compute the onset of instability (critical K)

Using the $\ell = 0$ principal condition:

$$\omega^* T + \tan^{-1}\left(\frac{\omega^*}{P}\right) = \pi, \quad K_{\text{crit}} = \sqrt{P^2 + (\omega^*)^2}.$$

For $T = 0.2$, $P = 1$, solving numerically gives approximately

$$\omega^* \approx 8.443, \quad K_{\text{crit}} \approx 8.502.$$

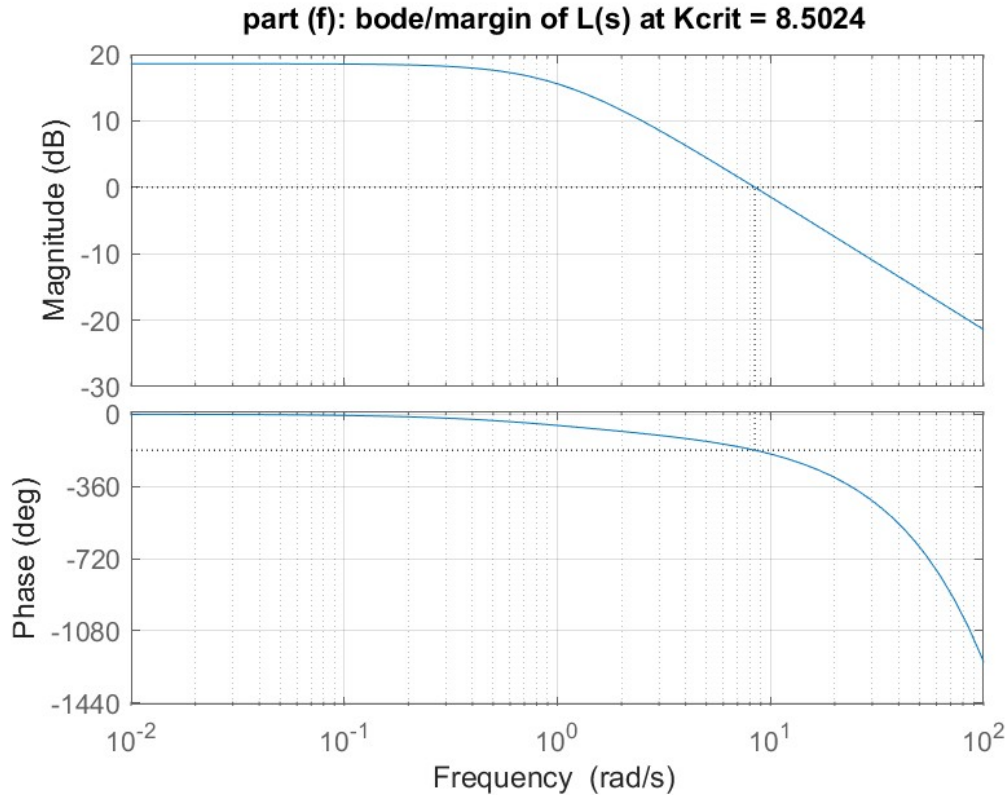
(Your MATLAB code below reproduces this via `fzero`.)

(f) What to expect in the Bode plot at $K = K_{\text{crit}}$

At marginal stability, $|L(j\omega^*)| = 1$ and $\angle L(j\omega^*) = -180^\circ$. Therefore:

$$\text{Phasemargin} \approx 0^\circ \text{ at } K = K_{\text{crit}}.$$

So the Bode magnitude crosses 0 dB at ω^* , exactly where the phase hits -180° .



(g) Step response at K_{crit}

For unity feedback, the closed-loop transfer is

$$T(s) = \frac{L(s)}{1 + L(s)}.$$

At the critical gain, you should observe an oscillatory response that is *barely damped* (on the verge of instability), consistent with phase margin $\approx 0^\circ$.

outputs : " Part (e): Critical (touching) values $w^* = 8.443413$ rad/s $K_{crit} = 8.502425$ $L(jw^*) = -1.000000 - 0.000000j$

Part (f): margins at K_{crit} Phase margin $PM = 0.0000$ deg Gain crossover W_{cp} ($|L|=1$) = 8.443413 rad/s phase crossover W_{cg} (phase=-180) = 8.443366 rad/s

Part (g): stepinfo at K_{crit} (interpret cautiously near marginal stability) overshoot = 485.67 SettlingTime = NaN s peakTime = 5299.1007 s"

Question 8:

In this question the loop is

$$L(s) = \frac{e^{-Ts}}{s + P},$$

and we consider a proportional gain K with unity negative feedback, so the open-loop is $KL(s)$. The closed-loop characteristic equation is

$$1 + KL(s) = 0 \iff 1 + \frac{Ke^{-Ts}}{s+P} = 0 \iff (s+P) + Ke^{-Ts} = 0.$$

Because of e^{-Ts} , this is transcendental and (in principle) has infinitely many roots.

(a) Condition for a point s to lie on the root-locus

A point $s \in \mathbb{C}$ belongs to the root-locus of $KL(s)$ if there exists a real $K \geq 0$ such that

$$1 + KL(s) = 0 \iff KL(s) = -1.$$

Therefore the standard root-locus magnitude/angle conditions still apply:

Angle condition.

$$\angle L(s) = (2\ell + 1)\pi, \quad \ell \in \mathbb{Z}.$$

Here

$$\angle L(s) = \angle(e^{-Ts}) - \angle(s+P).$$

Write $s = \sigma + j\omega$. Then

$$e^{-Ts} = e^{-T(\sigma+j\omega)} = e^{-T\sigma} e^{-jT\omega} \Rightarrow \angle(e^{-Ts}) = -T\omega.$$

Hence the angle condition becomes

$$-T\omega - \angle(\sigma + P + j\omega) = (2\ell + 1)\pi.$$

Magnitude condition.

$$|KL(s)| = 1 \iff K = \frac{1}{|L(s)|}.$$

But

$$|L(s)| = \frac{|e^{-Ts}|}{|s+P|} = \frac{e^{-T\sigma}}{|s+P|},$$

since $|e^{-Ts}| = e^{-T\sigma}$. Therefore

$$K = |s+P| e^{T\sigma}.$$

Together, the angle + magnitude constraints characterize the (infinite-dimensional) root-locus.

(b) Closed-loop poles for a given K (Lambert W closed form)

Starting from the characteristic equation

$$(s + P) + Ke^{-Ts} = 0 \iff (s + P)e^{Ts} = -K,$$

multiply and divide to isolate a Lambert W form:

$$(s + P)e^{T(s+P)} = -Ke^{PT}.$$

Let $y = T(s + P)$. Then

$$ye^y = -KTe^{PT}.$$

By definition of the Lambert W function (all branches),

$$y = W_k(-KTe^{PT}), \quad k \in \mathbb{Z}.$$

Hence the infinitely many closed-loop poles are

$$s_k(K) = -P + \frac{1}{T} W_k(-KTe^{PT}), \quad k \in \mathbb{Z}.$$

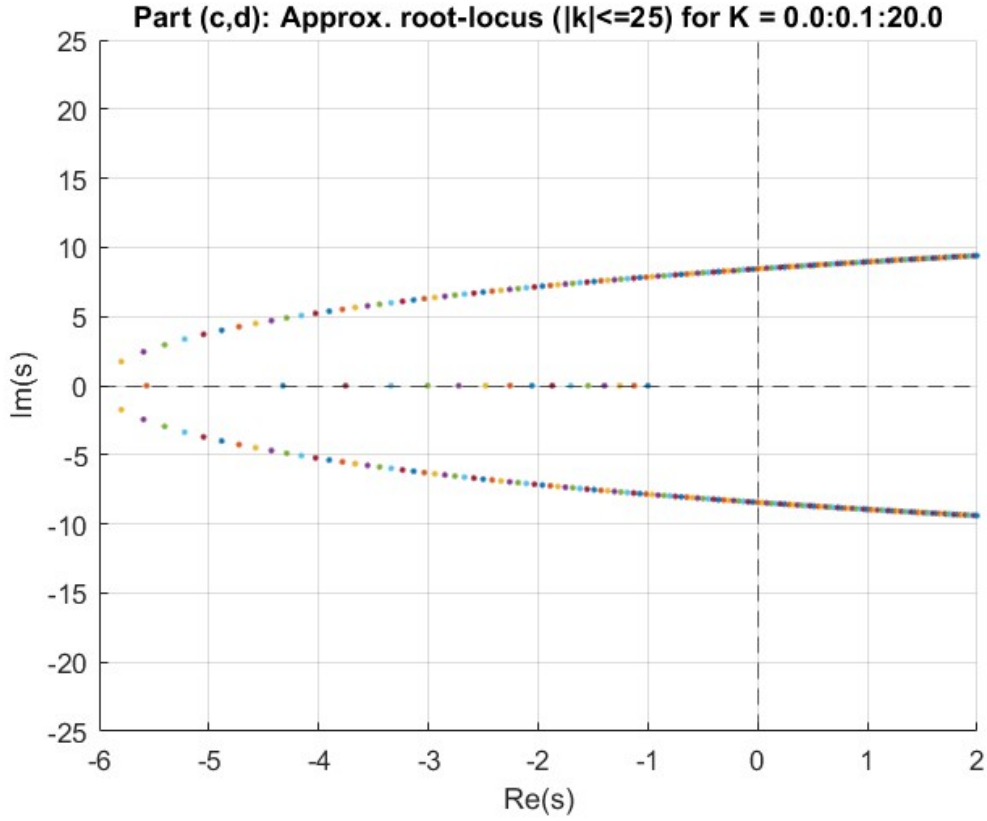
This is the key reason time-delay systems can produce multiple (indeed infinitely many) roots.

(c) Sweeping K and computing poles

Compute $s_k(K)$ over a range of gains $K \in [0, 20]$ and branches $k = -N, \dots, N$ (finite truncation), then plot all computed poles to approximate the root-locus.

(d) Sketching and the “multiple roots” phenomenon

As K varies, each branch k traces a locus in the s -plane. Because there are infinitely many branches of $W_k(\cdot)$, the system can have infinitely many closed-loop poles. A practical plot uses a finite number of branches $|k| \leq N$ and shows the dominant ones (closest to the imaginary axis).



(e) Gain that causes instability

Instability onset occurs when the rightmost pole crosses the imaginary axis, i.e. $\Re\{s\} = 0$ for some pole. Equivalently at marginal stability $s = j\omega^*$ satisfies

$$(j\omega^* + P) + Ke^{-j\omega^*T} = 0.$$

Taking magnitudes gives

$$K = \sqrt{P^2 + (\omega^*)^2}.$$

Taking angles (principal crossing for first instability onset) gives

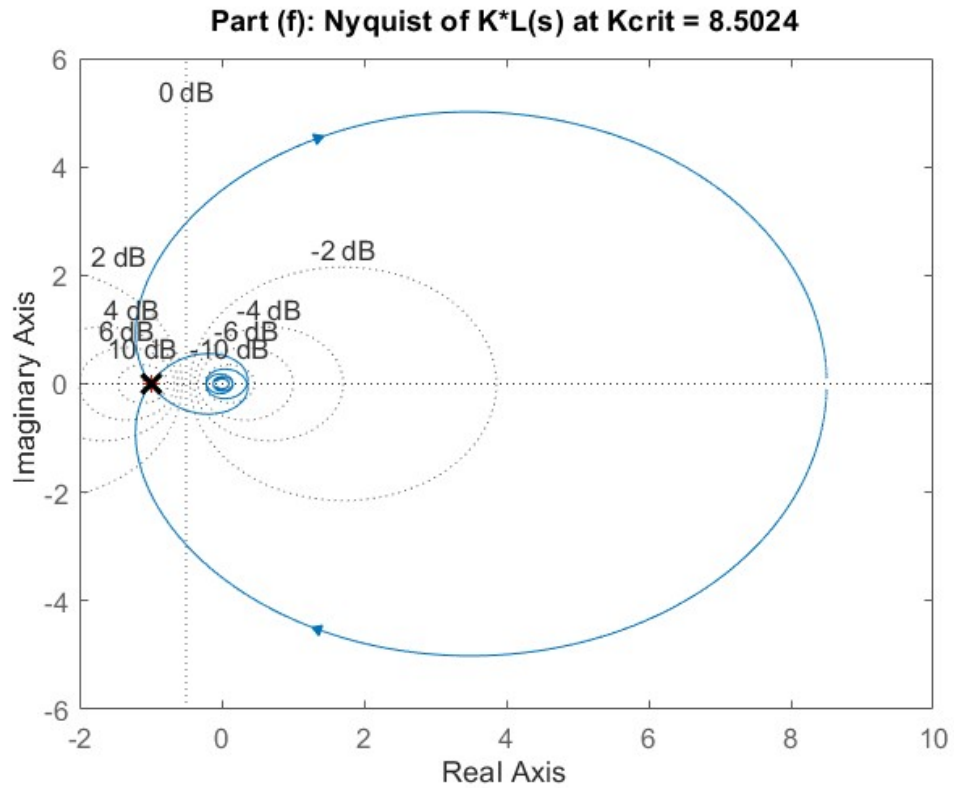
$$\omega^*T + \tan^{-1}\left(\frac{\omega^*}{P}\right) = \pi.$$

For $T = 0.2$, $P = 1$, this yields approximately

$$\omega^* \approx 8.443, \quad K_{crit} \approx 8.502.$$

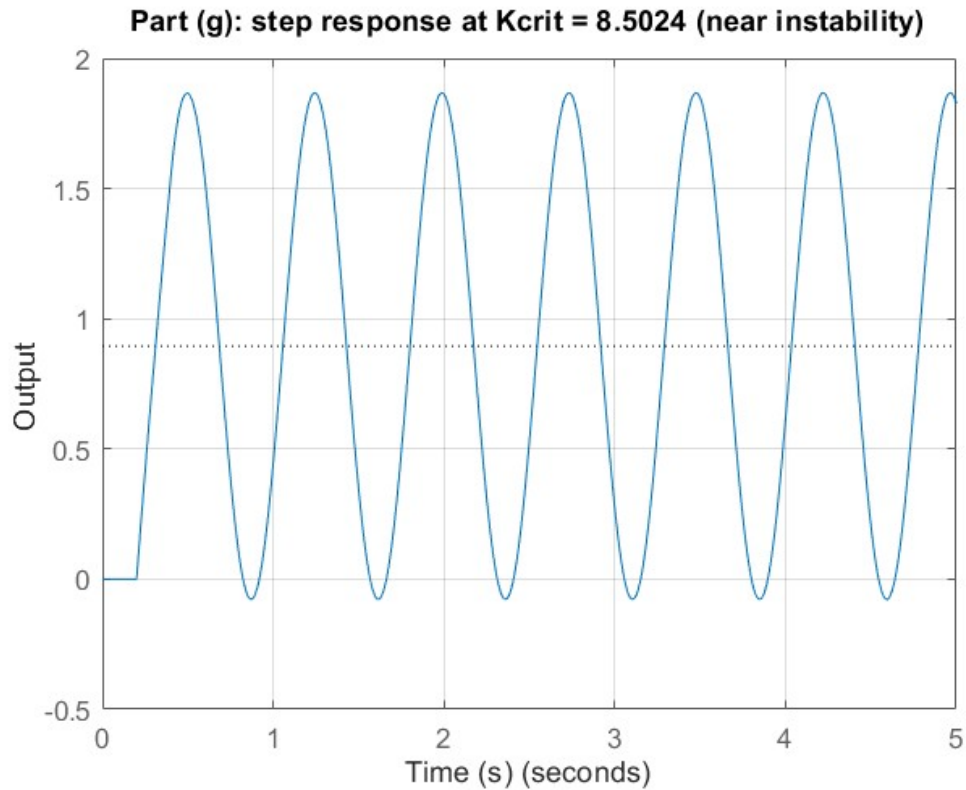
(f) Expectation from Nyquist at $K = K_{crit}$

At onset of instability the Nyquist plot of $KL(j\omega)$ touches the critical point -1 (marginal stability). So for $K = K_{crit}$, Nyquist should pass through $-1 + j0$; for $K > K_{crit}$, it begins encircling -1 .



(g) Step response at K_{crit}

With unity feedback, $T(s) = \frac{KL(s)}{1+KL(s)}$. At $K = K_{crit}$, the step response should show oscillations with (approximately) zero damping — “on the verge of instability.”



outputs : ” Part (b): Example $K = 5.00$, showing a few poles (truncated branches): $1.0e+02 *$

$-0.2513 - 7.6168i$ $-0.2492 - 7.3026i$ $-0.2470 - 6.9883i$ $-0.2447 - 6.6741i$ $-0.2423 - 6.3599i$ $-0.2398 - 6.0457i$ $-0.2371 - 5.7314i$ $-0.2343 - 5.4172i$ $-0.2313 - 5.1029i$ $-0.2282 - 4.7887i$

Part (e): Critical gain (first onset of instability) $w^* = 8.443413$ rad/s $K_{crit} = 8.502425$ $\max(\text{Re}(\text{poles}))$ at K_{crit} (truncated) = $-1.110223e-16$

Part (g): stepinfo at K_{crit} (interpret cautiously) Overshoot = 485.67 SettlingTime = NaN s PeakTime = 5299.1007 s $\ddot{u}$ ”